

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples (teacher)

1. Column 1, E1: $16 \rightarrow -16$
2. Column 1, E2: $7 \rightarrow -7$
3. Column 1, E3: 63 floors up $\rightarrow$ 63
4. Column 2, E1: $-(-26) \rightarrow$ = the opposite of -26 ; $= 26$
5. Column 2, E2: $-(+10) \rightarrow$ = the opposite of 10; $= -10$
6. Column 3, E1: The opposite of a number is -16 . What is the number? $\rightarrow$ opposite of -16 ; the number is 16
7. Column 3, E2: A diver is 27 m below sea level. Write this as an integer, then write its opposite. $\rightarrow$ below sea level: -27 ; opposite: 27
8. Column 3, E3: List all the integers whose distance from 0 is less than 4. $\rightarrow$ from -3 up to 3; $-3, -2, -1, 0, 1, 2, 3$
9. Column 3, E4: If $n = -32$, find $-n$ and $-(-n)$. $\rightarrow -n = -(-32) = 32$; $-(-n) = -32$

Questions (in order: Easy, Medium, Challenging)

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|-----------|-----------|------------------|
| 1. 59 | 13. 9 | 25. 49 |
| 2. -41 | 14. -29 | 26. -7 |
| 3. -9 | 15. -8 | 27. 57 |
| 4. -47 | 16. -48 | 28. -2 |
| 5. 25 | 17. -50 | 29. 25 |
| 6. -54 | 18. 52 | 30. -26 |
| 7. 14 | 19. 4 | 31. 29 |
| 8. 52 | 20. -22 | 32. 34 |
| 9. 43 | 21. -9 | 33. $-16, 16$ |
| 10. -23 | 22. -48 | 34. $-1, 0, 1$ |
| 11. -27 | 23. 16 | 35. 22 and -22 |
| 12. 21 | 24. -36 | 36. 3 |

1.02 The number line (booklet p. 4)

WE DO examples (teacher)

1. Column 1, E1: 5 or $-4 \rightarrow 5$
2. Column 1, E2: -2 or $-3 \rightarrow -2$
3. Column 1, E3: start at -1 , move 7 right $\rightarrow 6$
4. Column 2, E1: -7 and 11 $\rightarrow -7$ to 0 is 7, 0 to 11 is 11; $7 + 11 = 18$
5. Column 2, E2: -8 and 3 $\rightarrow -8$ to 0 is 8, 0 to 3 is 3; $8 + 3 = 11$
6. Column 2, E3: -10 and $-6 \rightarrow 4$ apart, half is 2; $-10 + 2 = -8$
7. Column 3, E1: How many integers are between -29 and 8? $\rightarrow -28$ to -1 : 28 integers, and 0; 1 to 7: 7 integers; $28 + 1 + 7 = 36$
8. Column 3, E2: Which integer is halfway between -24 and 0? $\rightarrow$ distance apart: 24; half: 12; $-24 + 12 = -12$

Questions (in order: Easy, Medium, Challenging)

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|---------|----------|--------|
| 1. 2 | 11. 10 | 21. 25 |
| 2. 8 | 12. 3 | 22. 16 |
| 3. 7 | 13. 15 | 23. 20 |
| 4. 1 | 14. 6 | 24. 3 |
| 5. 14 | 15. -9 | 25. 26 |
| 6. -2 | 16. 10 | 26. 0 |
| 7. 11 | 17. 3 | 27. 30 |
| 8. 14 | 18. 1 | 28. 0 |
| 9. -7 | 19. 4 | 29. 48 |
| 10. 3 | 20. 23 | |

1.03 Comparing and ordering integers

(booklet p. 5)

WE DO examples (teacher)

1. Column 1, E1: $-7 \square -8 \rightarrow >$

2. Column 1, E2: $-5 \square -10 \rightarrow >$
3. Column 1, E3: -7 or $-2 \rightarrow -7$
4. Column 2, E1: 15, -8 , -22 , $-9 \rightarrow$ smallest: -22 , largest: 15; $-22, -9, -8, 15$
5. Column 2, E2: 5, 14, -24 , $-18 \rightarrow$ smallest: -24 , largest: 14; $-24, -18, 5, 14$
6. Column 2, E3: 12, -27 , 25, $-32 \rightarrow$ largest: 25, smallest: -32 ; 25, 12, $-27, -32$
7. Column 3, E1: Write three integers greater than -14 but less than -9 . $\rightarrow$ integers from -13 to -10 ; e.g. $-13, -12, -11$
8. Column 3, E2: Which is the smallest: $-(-7)$, -7 or $-(-12)$? $\rightarrow -(-7) = 7$, $-(-12) = 12$; smallest: -7
9. Column 3, E3: Write in ascending order: $-(-2)$, -9 , -12 , 0 $\rightarrow -(-2) = 2$; $-12, -9, 0, 2$

Questions (in order: Easy, Medium, Challenging)

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|---------|----------------------|--------------------------|
| 1. $>$ | 13. $<$ | 25. $-8, -7, 4, 10$ |
| 2. $<$ | 14. $>$ | 26. $-23, -14, 4, 15$ |
| 3. $>$ | 15. -14 | 27. $-25, -16, 6, 20$ |
| 4. $<$ | 16. -14 | 28. 9, $-13, -16, -22$ |
| 5. $>$ | 17. -12 | 29. e.g. $-17, -16, -15$ |
| 6. $<$ | 18. -21 | 30. -8 |
| 7. $>$ | 19. -26 | 31. $-7, -3, 0, 4$ |
| 8. $>$ | 20. -24 | 32. e.g. $-18, -17, -16$ |
| 9. $>$ | 21. -14 | 33. -6 |
| 10. $<$ | 22. -30 | 34. $-11, -7, 0, 9$ |
| 11. $<$ | 23. $-20, 0, 3, 5$ | |
| 12. $<$ | 24. $-24, -6, 5, 15$ | |

1.04 Adding integers (booklet p. 6)

WE DO examples (teacher)

1. Column 1, E1: $-4 + 8 \rightarrow 4$
2. Column 1, E2: $-7 + 10 \rightarrow 3$
3. Column 1, E3: $8 + (-1) \rightarrow 7$
4. Column 2, E1: $-12 + (-11) + (-13) \rightarrow -23 + (-13); = -36$
5. Column 2, E2: $-4 + (-3) + (-6) \rightarrow -7 + (-6); = -13$
6. Column 3, E1: The temperature is -7°C and rises 8°C . What is the new temperature? $\rightarrow -7 + 8; = 1^{\circ}\text{C}$
7. Column 3, E2: A lift at level -3 goes up 6 floors. Which level is it on? $\rightarrow -3 + 6; =$ level 3
8. Column 3, E3: Sam has \$35 and spends \$51 on credit. What is the balance? $\rightarrow 35 + (-51); = -\16
9. Column 3, E4: Find the missing number: $-12 + \square = 6 \rightarrow -12$ to 6: 18 right; $\square = 18$

Questions (in order: Easy, Medium, Challenging)

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|----------|-----------|--------------------------|
| 1. -6 | 13. -3 | 25. 17 |
| 2. 6 | 14. -1 | 26. -10 |
| 3. -1 | 15. 10 | 27. -10 |
| 4. -2 | 16. -13 | 28. -2 |
| 5. -9 | 17. -1 | 29. 24 |
| 6. -1 | 18. -16 | 30. 7 |
| 7. 2 | 19. 7 | 31. 3 |
| 8. -6 | 20. 10 | 32. 10°C |
| 9. 1 | 21. 9 | 33. level 3 |
| 10. -4 | 22. 6 | 34. $-\$28$ |
| 11. 0 | 23. 5 | 35. 7 |
| 12. 0 | 24. -1 | |

1.05 Subtracting integers (booklet p. 7)

WE DO examples (teacher)

1. Column 1, E1: $-6 - 4 \rightarrow -10$
2. Column 1, E2: $-5 - 4 \rightarrow -9$
3. Column 1, E3: $12 - (-9) \rightarrow 21$

- Column 2, E1: $1 - (-15) + (-15) \rightarrow = 1 + 15 + (-15); = 16 + (-15); = 1$
- Column 2, E2: $7 - (-13) + (-6) \rightarrow = 7 + 13 + (-6); = 20 + (-6); = 14$
- Column 3, E1: The minimum temperature was -5°C and the maximum was 19°C . By how much did it rise? $\rightarrow 19 - (-5); = 19 + 5; = 24^{\circ}\text{C}$
- Column 3, E2: A submarine at -62 m rises 22 m . Where is it now? $\rightarrow -62 + 22; = -40\text{ m}$
- Column 3, E3: What is 2 less than -9 ? $\rightarrow -9 - 2; = -11$
- Column 3, E4: Find the difference between -6 m and 24 m . $\rightarrow 24 - (-6); = 24 + 6; = 30\text{ m}$

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|--------------------------|
| 1. -15 | 12. 6 | 23. -13 |
| 2. -23 | 13. -18 | 24. 11 |
| 3. 0 | 14. -9 | 25. 35 |
| 4. -10 | 15. 9 | 26. 15 |
| 5. -8 | 16. 11 | 27. 10 |
| 6. -3 | 17. 16 | 28. 9 |
| 7. -4 | 18. 11 | 29. 21°C |
| 8. -2 | 19. -7 | 30. -83 m |
| 9. -6 | 20. 6 | 31. -13 |
| 10. -8 | 21. 21 | 32. 33 m |
| 11. -10 | 22. 4 | |

1.06 Multiplying integers (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: $-5 \times 3 \rightarrow -15$
- Column 1, E2: $6 \times (-2) \rightarrow -12$
- Column 1, E3: $2 \times (-7) \rightarrow -$
- Column 2, E1: $4 \times (-5) \times (-3) \rightarrow = -20 \times (-3); = 60$
- Column 2, E2: $1 \times (-1) \times 4 \rightarrow = -1 \times 4; = -4$
- Column 3, E1: A diver goes down 2 m every minute. Where is she after 3 minutes? $\rightarrow -2 \times 3; = -6\text{ m}$
- Column 3, E2: The temperature drops 4°C each hour for 3 hours. What is the total change? $\rightarrow -4 \times 3; = -12^{\circ}\text{C}$
- Column 3, E3: Find the missing number: $-4 \times \square = -12 \rightarrow \square = -12 \div (-4); \square = 3$

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|---------------------------|
| 1. 63 | 16. + | 31. -100 |
| 2. -49 | 17. + | 32. 18 |
| 3. 3 | 18. - | 33. -100 |
| 4. -3 | 19. - | 34. -20 |
| 5. 7 | 20. + | 35. 16 |
| 6. -72 | 21. - | 36. 5 |
| 7. -36 | 22. - | 37. 24 |
| 8. -64 | 23. -36 | 38. -15 |
| 9. -80 | 24. 10 | 39. -8 |
| 10. -5 | 25. 150 | 40. 120 |
| 11. 56 | 26. -1 | 41. -12 m |
| 12. -36 | 27. -60 | 42. -36°C |
| 13. 40 | 28. -12 | 43. -8 |
| 14. -56 | 29. -18 | 44. -28 m |
| 15. - | 30. -2 | 45. -45°C |

1.07 Dividing integers (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $36 \div (-4) \rightarrow -9$
- Column 1, E2: $-28 \div (-7) \rightarrow 4$
- Column 1, E3: $30 \div \square = -5 \rightarrow -6$
- Column 2, E1: $8 \div 4 \div 2 \rightarrow = 2 \div 2; = 1$
- Column 2, E2: $-40 \div (-5) \div (-2) \rightarrow = 8 \div (-2); = -4$
- Column 3, E1: A debt of $\$160$ is shared equally by 8 people. Write each share as an integer. $\rightarrow -160 \div 8; = -\20 each

- Column 3, E2: The temperature fell 15°C over 3 hours. What was the change each hour? $\rightarrow -15 \div 3; = -5^{\circ}\text{C}$ each hour
- Column 3, E3: Find the missing number: $\square \div 8 = 4 \rightarrow \square = 4 \times 8; \square = 32$

Questions (in order: Easy, Medium, Challenging)

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|---------|--------|--------------------------|
| 1. -1 | 16. 1 | 31. 3 |
| 2. -10 | 17. 5 | 32. -5 |
| 3. 4 | 18. -2 | 33. 1 |
| 4. -5 | 19. -1 | 34. -2 |
| 5. -4 | 20. 1 | 35. 2 |
| 6. 8 | 21. -1 | 36. 5 |
| 7. -8 | 22. -5 | 37. 3 |
| 8. -8 | 23. -5 | 38. -3 |
| 9. 3 | 24. -3 | 39. 2 |
| 10. -4 | 25. 2 | 40. 2 |
| 11. 1 | 26. -1 | 41. $-\$19$ |
| 12. -10 | 27. 2 | 42. -2°C |
| 13. -4 | 28. 1 | 43. 20 |
| 14. -3 | 29. -5 | 44. $-\$18$ |
| 15. 9 | 30. -2 | 45. -6°C |

1.08 Order of operations with integers

(booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: $3 + (-6) \times (-6) \rightarrow 39$
- Column 1, E2: $-9 - 5 \times (-6) \rightarrow 21$
- Column 1, E3: $(-6 - (-1)) \times 2 \rightarrow -10$
- Column 2, E1: $6 \times 1 + 2 \div 1 \rightarrow = 6 + 2; = 8$
- Column 2, E2: $3 \times (-6) + (-16) \div 4 \rightarrow = -18 + (-4); = -22$
- Column 3, E1: $7 - (-4 + 6) \times (-3) \rightarrow = 7 - 2 \times (-3); = 7 - (-6); = 13$
- Column 3, E2: $6 - (-1 + (-2)) \times (-1) \rightarrow = 6 - (-3) \times (-1); = 6 - 3; = 3$

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|---------|
| 1. -29 | 14. 11 | 27. 7 |
| 2. -16 | 15. -21 | 28. -26 |
| 3. -36 | 16. -48 | 29. -17 |
| 4. 34 | 17. 9 | 30. 14 |
| 5. 22 | 18. -2 | 31. -34 |
| 6. 13 | 19. 13 | 32. 34 |
| 7. 19 | 20. -32 | 33. 15 |
| 8. 13 | 21. 5 | 34. 34 |
| 9. 19 | 22. 4 | 35. 14 |
| 10. 17 | 23. 29 | 36. -14 |
| 11. -22 | 24. -3 | 37. -4 |
| 12. -22 | 25. 1 | |
| 13. 8 | 26. 1 | |

1.09 Integers on a calculator (booklet p. 11)

WE DO examples (teacher)

- Column 1, E1: $252 + (-391) \rightarrow -139$
- Column 1, E2: $-369 + 350 \rightarrow -19$
- Column 1, E3: $43 \times (-4) \rightarrow -172$
- Column 2, E1: $(-86 - (-81)) \times (-16) \rightarrow \text{bracket: } -5; -5 \times (-16) = 80$
- Column 2, E2: $(-3 - 67) \times (-11) \rightarrow \text{bracket: } -70; -70 \times (-11) = 770$
- Column 3, E1: Estimate -60×44 , then calculate. $\rightarrow$ estimate: $-60 \times 40 \approx -2400$; calculator: -2640
- Column 3, E2: Estimate -32×48 , then calculate. $\rightarrow$ estimate: $-30 \times 50 \approx -1500$; calculator: -1536

Questions (in order: Easy, Medium, Challenging)

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|----------|------------|----------------------------|
| 1. 44 | 15. 66 | 29. 1 245 |
| 2. -284 | 16. -1 980 | 30. -390 |
| 3. 130 | 17. -33 | 31. -48 |
| 4. -121 | 18. 900 | 32. ≈ -600 ; -578 |
| 5. -14 | 19. -264 | 33. $\approx -2\ 400$; -2 |
| 6. -5 | 20. -561 | 088 |
| 7. 413 | 21. -992 | 34. $\approx -1\ 600$; -1 |
| 8. 66 | 22. -429 | 470 |
| 9. -155 | 23. 1 716 | 35. $\approx -1\ 500$; -1 |
| 10. 91 | 24. -190 | 288 |
| 11. -207 | 25. -1 278 | 36. $\approx -1\ 200$; -1 |
| 12. -330 | 26. 2 226 | 140 |
| 13. 308 | 27. 140 | 37. ≈ -600 ; -741 |
| 14. 324 | 28. 2 831 | 38. $\approx -1\ 500$; -1 |
| | | 248 |

1.10 Integer problems (booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: 5°C , rises $14^{\circ}\text{C} \rightarrow 19^{\circ}\text{C}$
- Column 1, E2: 14°C , falls $4^{\circ}\text{C} \rightarrow 10^{\circ}\text{C}$
- Column 1, E3: $-\$72$, deposit $\$92 \rightarrow \20
- Column 2, E1: a hill 267 m above sea level and a reef 57 m below $\rightarrow 267 - (-57); = 324\text{ m}$
- Column 2, E2: a hill 280 m above sea level and a reef 74 m below $\rightarrow 280 - (-74); = 354\text{ m}$
- Column 3, E1: A diver is at -45 m . She rises 4 m a minute for 4 minutes. Where is she now? $\rightarrow 4 \times 4 = 16; -45 + 16; = -29\text{ m}$
- Column 3, E2: A game scores +4 for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score? $\rightarrow 3 \times 4 = 12; 5 \times (-3) = -15; 12 + (-15) = -3$
- Column 3, E3: It is -2°C at 6 a.m. It warms 3°C an hour for 3 hours. What is the temperature? $\rightarrow 3 \times 3 = 9; -2 + 9; = 7^{\circ}\text{C}$

Questions (in order: Easy, Medium, Challenging)

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|---------------------------|---------------------------|--------------------------|
| 1. -21°C | 12. -4°C | 23. 116 m |
| 2. -23°C | 13. -24°C | 24. 85 m |
| 3. 6°C | 14. 26°C | 25. 84 m |
| 4. 4°C | 15. $-\$112$ | 26. 222 m |
| 5. 2°C | 16. $\$1$ | 27. 445 m |
| 6. 0°C | 17. $\$203$ | 28. 174 m |
| 7. -6°C | 18. $\$171$ | 29. 116 m |
| 8. -17°C | 19. $-\$91$ | 30. -5 m |
| 9. 21°C | 20. $\$25$ | 31. 12 |
| 10. -1°C | 21. $\$77$ | 32. 27°C |
| 11. -14°C | 22. $\$55$ | |